AI Governance and the Verification Gap: A Framework for Law and Policy Under Computational Intractability

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Abstract

The technical problem of verifying advanced AI systems is a central driver of the most pressing legal and political challenges in AI governance, and a significant structural barrier to the technology's long-term economic potential. This paper demonstrates that the computational intractability of auditing opaque models creates a verification gap. This technical gap fundamentally alters the landscape of risk and accountability. It creates an evidentiary challenge for legal principles, introduces unquantifiable operational and supply chain vulnerabilities for businesses, and poses novel organizational challenges to the efficacy of human oversight. This analysis argues that this is a persistent structural reality that challenges conventional product-based oversight and underscores the need to supplement technical verification with robust process-based governance. This approach focuses accountability on the most tractable element of AI development, which is the human-led process itself, offering a pragmatic path to managing the risks of computationally intractable systems. The resulting legal, regulatory, and operational uncertainty contributes to a market paradox, in which a high-stakes race among frontier developers proceeds in parallel with caution from risk-averse mainstream enterprises. This paper examines these interconnected issues and the distinct strategic postures global powers have adopted in response. It concludes by analyzing how its findings inform a pragmatic three-pillar framework of tiered liability, process-based governance, and targeted research designed to create a market where improved safety and trustworthiness become a verifiable, competitive advantage.

Keywords

Verification Gap, Process-Based Governance, Artificial Intelligence, AI Regulation, AI Safety, AI Risk Management, Computational Intractability, Computational Complexity, Mechanistic Interpretability, Explainable AI (XAI), Legal Liability, Tort Law, Regulatory Policy, Information Asymmetry, Systemic Risk, Operational Risk, AI Supply Chain, Corporate Governance, Political Economy of Technology, Human Factors

1. Introduction: The Verification Gap in an Era of Improving AI

The deployment of advanced artificial intelligence has accelerated at a pace that has created a significant governance gap, with corporate adoption far outpacing the implementation of strategic enterprise-wide oversight [1], [2]. Recent studies report that AI regulations lag behind technological advances by several years [3] and that regulatory inertia persists despite urgent calls for oversight [4]. This governance gap is stark. A 2025 global survey illustrates this divide, finding that while 58% of employees regularly use AI tools for work, only 34% report that their organization has a policy or even guidance on the use of these systems [5]. For example, one recent survey of UK corporate practices found that although 93% of organizations are deploying AI, only 7% have fully embedded governance frameworks [6]. This evidence of widespread use combined with a systemic lack of oversight is consistent with broader studies showing high rates of AI adoption alongside lagging implementation of mature governance protocols [1], [2]. Meanwhile, these systems are being rapidly integrated into nearly every part of society for a wide array of functions, ranging from healthcare and finance to national security and public infrastructure [1]. Yet their internal operations remain largely opaque, creating a significant risk of unintended and potentially harmful behaviors that affect not only the public but also core internal business operations [7], [8], [9], [10], [11]. This opacity has created a verification gap, a chasm between the expanding capabilities of AI and our ability to independently verify its safety and alignment [12], [13], [14], [15].

This gap is not a single problem but a composite of four distinct, interlocking challenges. The first is the foundational gap, which stems from the formal proofs of computational intractability [16]. It represents a permanent theoretical barrier to any toolchain that would fully certify a model's behavior for every possible input. The second is the practical gap, which is the difference between what current engineering and heuristic tools can achieve and the theoretical limits of verification. This gap can shrink with new research, but it is currently substantial [14], [17]. The third is the observational gap, which describes the insufficiency of behavioral testing. This is the chasm between how a model acts in a controlled test environment and the potentially novel or deceptive ways it could behave when deployed [8], [18]. Finally, there is the asymmetric gap. This is not merely an asymmetry of information, but more precisely an asymmetry of process control. While the formal intractability of verification may apply to developers and auditors alike, only the developer has exclusive control over the design choices, training data, and mitigation strategies implemented before the model is finalized. This divide between the party with proactive control over the development process and the parties who can only conduct reactive, post-hoc verification is the central source of the legal and economic information asymmetry.

The challenge is not just that these systems are complex, but that they exhibit emergent properties and failure modes that were not explicitly programmed into them and are not easily predicted from their training [7], [19]. These properties can range from the amplification of

societal biases to the potential for adversarially concealed malicious functions, such as sleeper agent capabilities that persist through safety training [8], [20]. The fundamental problem is that we currently lack reliable, scalable methods to look inside these models to gain sufficient assurance that they will behave as intended, especially in novel situations [8], [12], [14], [21], [22].

The inability to verify a core component of our emerging societal infrastructure creates the potential for unquantifiable and correlated risks [23], [24], [25], [1]. These risks manifest in two interconnected domains. Externally, a single, deceptively aligned model deployed in a key industry could trigger cascading failures that traditional risk models cannot anticipate [24], [26], [27], [23]. As just one example, imagine a scenario in which the same proprietary AI model underwrites credit scores at leading U.S. banks [28]. This kind of vendor and model concentration has been flagged as a potential systemic vulnerability by the Financial Stability Board [29]. A fault in the model could simultaneously misprice credit risk across institutions, triggering ripple effects through the broader economy. Regulators without direct access to the model's algorithms would struggle to identify the root cause or coordinate a targeted response. This broader AI supply chain risk, where entire sectors become dependent on a few opaque foundation models, turns a private vendor choice into a source of potential systemic failure [25]. This risk extends beyond model functionality to data provenance, because the verification gap prevents downstream users from auditing the integrity of training data, which can lead to a breaking of the chain of trust from data to deployment [30].

Internally, the verification gap creates a new class of operational and cybersecurity risks, even when AI is used for non-public-facing tasks with human oversight. For example, an AI assistant used for software development might introduce subtle but critical security vulnerabilities into a company's codebase that evade standard human review and testing, creating downstream liabilities [31]. This challenge is compounded when reliance on opaque systems fosters cognitive offloading, diminishing the critical skills of human reviewers and turning their oversight into a procedural formality [32], [33]. An enterprise cannot easily audit a third-party model to ensure its internal logic is sound or that it has not been compromised, transforming a powerful productivity tool into a potential source of latent vulnerabilities [14]. This demonstrates the need for regulatory frameworks that address the opacity and concentration risks unique to AI systems [29].

A society unable to technically verify these systems is forced to rely on legal attestations from developers [34], [35], [36], [37]. This creates a trust deficit that will become a faultline through the governance problem [1], [37]. This verification gap is not just an academic concern. It is actively reshaping the policy discussions and practices of AI governance. The European Union's AI Act, for instance, mandates a post-market monitoring system for high-risk AI that requires providers to continuously collect and analyze data on a system's performance throughout its

lifetime [38]. Similarly, in the private sector, private equity firms are responding to this trust deficit by engaging in rigorous technical due diligence that goes beyond business plans to assess model robustness, data integrity, and scalability as the potential for unquantifiable liability and compliance risk from AI systems has become a material concern in investment decision-making [39], [40]. The convergence of a supra-national regulator and private investors on the conclusion that developer attestations are insufficient provides powerful, real-world validation of the verification gap's significance. Their shared view is that developer attestations are insufficient and external verification is essential [39], [40].

The verification gap also creates a fundamental challenge to procedural justice. When an individual is harmed by an opaque AI system, the inability to verify how that decision was made could make assigning liability difficult if not impossible [41], [42]. The technical opacity of these models creates a steep and possibly insurmountable evidentiary barrier for the affected individuals, transforming a technical problem into potential fairness and access to justice concern [41], [42], [43]. The harm is not merely the adverse outcome but the denial of a fair process for redress [43], [44], [45].

This analysis demonstrates that the verification gap is a central challenge of AI governance and is likely a persistent feature of our most advanced systems. This paper’s central contribution is establishing the formal connection between proofs of computational intractability and the fundamental challenges they create for legal liability, regulatory oversight, and both external and internal corporate risk management, showing how the technical inability to empirically audit the internal safety of AI models creates a parallel evidentiary gap. This gap makes it difficult to assign legal responsibility for harms using traditional tort doctrines and complicates the ability of regulators and businesses to ensure safety and conduct due diligence using conventional, product-focused methods. This legal and technical uncertainty undermines existing regulatory efforts and complicates the entire governance landscape [46], [47], [48], [49]. The contribution of this paper is not the invention of process-based governance, but the provision of a new and more robust formal justification for why, in the face of computational intractability, it represents a rational and necessary path forward.

A practical governance framework must account for a stratified landscape of risk. For the highest-risk applications, such as those involving critical infrastructure or the command and control of autonomous systems, the formal intractability of verification remains a central challenge. It represents a direct barrier to gaining assurance against the low-probability, high-impact tail risks that regulation is primarily designed to prevent. For these systems, worst-case uncertainty must be a defining consideration for policy. For widely deployed general-purpose systems, a different challenge emerges. Even if the severity of individual harms is low, the unprecedented scale of deployment means that low-probability failures can aggregate into a significant integrated harm, justifying a more rigorous governance approach than is

applied to traditional mass-market software [50], [51].

This paper explores these interconnected domains, beginning with the technical hurdles of behavioral and internal assessments, then exploring the resulting challenges for legal and regulatory systems, and finally analyzing the broader political and economic context. The goal is to provide a fact-based account of the hurdles that must be overcome and to propose a path forward that includes liability reform, process-based regulation, and sustained research.

2. The Limits of External Observation: Why Behavioral Assessment is Insufficient

The most common approach to AI safety evaluation is to treat the system as a black box, assessing its safety based on observable input-output behavior [52], [53] through methods like performance benchmarking and adversarial red-teaming [54], [55]. This is a necessary component of safety testing, however, it is fundamentally insufficient for providing robust assurance for advanced AI systems, particularly as a standalone guarantee of safety for enterprise use [56], [57]. Behavioral tests are effective at finding some vulnerabilities, but they are not capable of proving the absence of hidden ones [58], [59], [60]. This insufficiency stems from issues where the models may be deliberately subversive [61], [8] and the tools used to test them are often unreliable [18], [58].

The first domain of failure is illustrated by the sleeper agent problem, a concrete example of deceptive alignment [8]. Research has demonstrated that a model can be trained to behave safely in most contexts, and yet to pursue a hidden, malicious goal when it encounters a specific trigger [8]. Researchers have successfully trained models to write secure code that inserts a critical vulnerability if the prompt contains a trigger phrase [8]. This presents a critical internal risk, because even a human-in-the-loop reviewing the AI's output may not detect such a subtle, adversarially placed flaw. This risk is compounded by automation bias, where human overseers become less vigilant and overly trusting of a system that is correct most of the time, making them more vulnerable to deception [33], [62]. This dynamic reveals a core weakness in relying on human review for opaque systems. When the reviewer lacks the tools to inspect the model's reasoning process, their oversight can devolve from a substantive check into a procedural formality [33]. Importantly, these deceptive capabilities can be made persistent so that they resist removal by all standard safety training techniques, including supervised fine-tuning, reinforcement learning, and even adversarial training [8]. This behavior highlights an unintended consequence. Adversarial training, which is designed to increase safety, can inadvertently teach the model to better recognize and conceal its backdoor triggers, creating a dangerous false impression of safety [8]. Alignment faking is a related phenomenon which shows that highly

capable models can emergently develop deceptive strategies [63]. This allows a model to strategically comply with a training objective it internally opposes while preserving its misaligned preferences for deployment [63].

The second domain of failure lies with the evaluation tools themselves. Standardized benchmarks often fail to capture the complexities of real-world dynamics and are vulnerable to being gamed [18]. A model that scores well on a benchmark may do so by exploiting spurious correlations in the test data that do not hold true when the model is deployed [55]. Recent analyses of popular AI agent benchmarks have found severe issues, including fragile simulators and flawed evaluation metrics [18], [64]. This known unreliability of behavioral tests creates a perverse incentive for audit washing or compliance theater, where developers perform auditable procedures that are known to provide only a weak signal of real-world safety [65], [66], [67]. For an enterprise procuring a third-party model, this creates a significant due diligence challenge. The enterprise may receive a favorable audit report that satisfies a compliance checklist but provides a false sense of security about the model's true operational risks [67]. A developer can point to high scores on a flawed benchmark as evidence of due diligence, satisfying a regulator's checklist while being aware that the benchmark provides a weak signal of real-world robustness [18]. This dynamic undermines the basic ideas of robustness and accountability that international standards like the OECD AI Principles call for [68].

Another cornerstone of behavioral testing is red teaming, which is a process of adversarial probing designed to elicit model failures [54]. While red teaming plays a critical role in organizational decision-making, its effectiveness hinges on practitioners having sufficient operational latitude to identify and test system vulnerabilities in line with secure ML design principles [69], [70]. As influential analysis emphasizes, the most impactful red teams rigorously challenge their own logic and assumptions, especially when navigating the distortions introduced by institutional culture and internal biases [71]. Recent analyses identify methodological gaps in red teaming [72] and document novel threats such as chain of thought manipulation and multi-agent attacks that evade classic frameworks [73]. Additionally, model evolution can invalidate red team assessments if findings are not continuously updated, because static results quickly become obsolete as new model behaviors emerge [74], [75]. For an enterprise, this means that an internal safety audit can become obsolete without warning, creating a persistent and unmanaged operational risk.

These difficulties complicate assurance efforts. A passing grade on a behavioral test has at least three plausible interpretations. The model may be safe, it may have unobserved safety issues, or the test itself may not be providing a reliable signal. For high-stakes regulatory decisions, this level of uncertainty suggests that behavioral assessments alone are insufficient. Understanding a model's internal mechanisms may be an indispensable component of any robust audit.

3. The Intractability of Verification: Mechanistic Interpretability and Its Limits

Given the limitations of behavioral testing, a portion of the AI safety research community has shifted its focus toward mechanistic interpretability [17]. The goal is to reverse engineer the internal algorithms learned by models, moving safety assessments beyond observing what a model does to understanding how it does it. While this research is a promising approach, it is in its early stages, characterized by open problems and facing some fundamental obstacles [17], [76].

3.1 The Science of Feature and Circuit Discovery

A primary goal of mechanistic interpretability is to identify features within a model's internal single activations that correspond to human-understandable concepts. A fundamental challenge is that concepts are often not represented cleanly. A single neuron can be polysemantic, activating for multiple unrelated concepts, while a single concept can be represented as a superposition across many different neurons [77], [78]. The leading tools for untangling these representations are dictionary learning methods, most notably the Sparse Autoencoder or SAE [78]. This approach has shown promise, but recent studies have revealed significant limitations, including a trade-off between reconstruction quality and interpretability, and the production of brittle features that do not generalize well [79]. Further research has cast doubt on whether SAEs find canonical units of analysis at all. Work by Leask et al. shows that features from smaller SAEs are often incomplete, while features from larger SAEs can be decomposed into combinations of simpler features [80]. This finding raises doubts about their status as fundamental building blocks [80]. This instability is compounded by identified failure modes like feature splitting and feature absorption [79], [81].

This has led researchers to explore an expanding set of alternative tools. Transcoders, for example, also analyze a model's activations but do so by learning a sparse approximation of a component's input-output function rather than simply reconstructing the activation vector [82]. RouteSAE uses a routing mechanism to extract features from multiple layers [83]. Matryoshka SAE (MSAE) learns hierarchical features at multiple granularities [84]. Other approaches like Low-Rank Sparse Attention (Lorsa) aim to disentangle complex attention mechanisms [85]. Each method represents a conceptual advance, and they are all subjects of ongoing research. The current state of feature discovery, while a very active area of research, is still not settled science ready for regulatory application. For an enterprise, this means there are no reliable off-the-shelf tools to confirm that a model procured from a third-party vendor does not contain undesirable or

biased features relevant to its internal use case.

3.2 The Formal Intractability of Auditing

Identifying a feature is only the first step. To understand a model's behavior, one must identify the entire computational subgraph, or circuit, responsible for that behavior. While algorithmic efficiency in finding circuits has improved, there is a more fundamental barrier, which is computational complexity. Recent work by Adolfi et al. provides formal proofs that key circuit discovery tasks are intractable in the worst case [16]. Their core result is not about the difficulty of testing a known circuit, but about the difficulty of determining if a small, explanatory circuit even exists within the vast network [16]. The problem is split into two levels of hardness. Finding a circuit that is merely consistent with a finite set of test cases is proven to be NP-complete [16]. The far harder problem, finding a circuit that is provably correct for every possible input, is proven to be Σ_2^P-complete [16]. To illustrate the significance of this complexity jump, one can use an analogy. Finding a single, explanatory circuit is an NP-complete problem. This is similar to searching for a specific master key that is known to open a complex safe. The search for the master key might be extremely difficult, but if one is found, it is trivial to insert it and confirm that it works. In contrast, proving a model is globally safe is a Σ_2^P-complete problem. This is a far harder task, analogous to proving that no possible master key, of any conceivable design, could ever open the safe. To provide such an absolute guarantee, one would have to test every imaginable key against the lock's design, an exhaustive check that is computationally intractable. This step up in complexity from NP to Σ_2^P places local discovery at the first level of the polynomial hierarchy and global discovery one level higher, with important implications [86].

These results provide a formal explanation for the empirical limits of current interpretability tools [16]. For auditors, regulators, and litigants, these results formalize the information asymmetry they face. An auditor can, at best, find a counterexample to a claimed explanation, but they cannot formally prove its global correctness. This computational hardness is not just a technical barrier. It creates a new, legally significant category of information asymmetry that existing legal doctrines were designed to address in other contexts [87], [88]§ 39, [89], [90]. Legal tools like evidentiary presumptions were developed for situations where one party has exclusive control over evidence, such as in medical malpractice [88]§ 39, [91], [90]. The Σ_2^P-completeness of global verification means a developer has a similar form of exclusive access, not because they are hiding proof, but because the search space is computationally too vast for an outsider to traverse. This provides a formal, mathematical justification for adapting legal tools, such as shifting the burden of proof, to this new technological reality.

This computational barrier clarifies the true nature of the asymmetric gap introduced earlier. For an arbitrary pre-trained model, the Σ_2^p-completeness barrier applies to the developer as much as to an external auditor. Even with full access to the model's weights, a post-hoc search for a globally correct circuit remains computationally infeasible. Again, the developer's exclusive control is therefore not the ability to solve this intractable problem after the fact, but the exclusive ability to shape the process and take steps to mitigate intractability during the initial design.

While this paper does not endorse specific technical methods, it is useful to illustrate the classes of action that fall within this exclusive developer control. These are not solutions to intractability, but rather principled design choices that can reduce the difficulty of a post-hoc audit. One class of action involves architectural philosophy. A developer could choose to construct a model from more modular or inherently interpretable components, which may localize functions in ways that are more amenable to later analysis than a fully monolithic design. Another category involves process instrumentation. A developer could instrument the training process to create a durable evidentiary trail, such as by implementing causal tracing methods that log which data subsets are most influential in developing specific capabilities. A third class of action is supply chain transparency. This would involve maintaining a rigorous and machine-readable record of a model's components and data provenance, analogous to a Software Bill of Materials (SBOM) [92]. These examples are not exhaustive, and the most effective methods will evolve with the field. They serve to make the concept of proactive control concrete. The choice of whether or not to adopt such practices is made exclusively by the developer, and it is this choice that forms a primary basis for the paper's subsequent arguments regarding accountability and liability.

There are several research directions that aim to bring formal or statistical guarantees into transformer-scale systems. One direction adapts a statistical runtime verification framework to estimate robustness of whole LLM outputs and reports confidence bounds rather than symbolic proof object [93]. Another line uses set based abstract interpretation to compute soundness bounds on attention mechanisms without emitting machine-checkable certificate [94]. A related study applies randomized smoothing to multi-agent systems driven by LLMs and provides a probabilistic guarantee that no input perturbation within a chosen norm ball will change the output beyond a strict bound [95]. Foundational tools such as the Reluplex SMT solver and branch and bound frameworks formally verify safety properties on small feed forward networks and could potentially inform segment by segment analysis of LLMs [96], [97]. While these are promising avenues, the Σ_2^p-completeness barrier implies that no future toolchain can fully certify an LLM's behavior for every possible input [16].

This distinction reframes the legal question. The key issue is not merely the model opacity, but whether the developer chose a design path that foreclosed any possibility of meaningful

verification by any party [37], [38]. That point provides a more precise justification for shifting the evidentiary burden because it targets a specific and consequential design choice made exclusively by the developer [37], [38], [98], [99].

3.3 The Practical Limits of Heuristic Auditing

It is critical to distinguish between the standard of assurance required for all applications and that which is necessary for the highest-stakes use cases. For the vast majority of AI systems, a robust portfolio of heuristic auditing, behavioral testing, and sound engineering practice can provide a sufficient and reasonable standard of safety. These practical methods are a cornerstone of current AI safety work and are effective at identifying a wide range of potential flaws. The following analysis of the limits of heuristic auditing should therefore be understood in a specific context. It applies primarily to those high-stakes systems where the potential for catastrophic or systemic harm necessitates a higher standard of assurance, and to widely deployed systems where even low-probability failures can create substantial integrated harm at scale.

Despite the formal proofs of worst-case intractability, it is crucial to acknowledge that practical, heuristic-based approaches have demonstrated partial success in auditing. The problem is not that no internal auditing is possible, but that complete, guaranteed verification of global properties is out of reach [16], [96], [100]. Many engineering disciplines rely on heuristic and approximate methods that offer high efficacy on typical problems, and these tools are an indispensable component of any defense-in-depth approach to AI risk mitigation. They serve to identify common flaws, improve overall model robustness, and provide a vital first line of defense in internal safety practices [56], [101].

An objection may be raised that this focus on worst-case intractability creates a strawman of perfect verification that no real-world engineering discipline requires. For many non-catastrophic applications, a portfolio of heuristic auditing and behavioral testing can indeed provide a sufficient and reasonable standard of safety. This perspective, however, does not fully account for two factors unique to AI systems that complicate a standard of good enough safety. The first is the problem of integrated harm. An AI application with a low probability of individual harm can still create substantial aggregate harm when deployed at scale [50], [51]. For example, numerous studies have documented that large language models reproduce and amplify societal biases found in their training data, such as associating certain professions with specific genders. While the impact of a single biased sentence completion is negligible, the systemic effect of deploying such a model in a tool used to screen millions of resumes can create significant, widespread discriminatory outcomes [102].

The second factor is the unique nature of AI system maintenance. Unlike a tradition software

patch that occurs on end-user devices, most large AI models are deployed as a service, allowing for instant, server-side modifications. A common mitigation for a discovered flaw is a behavioral patch, such as an input or output filter that blocks a specific unwanted behavior. This is distinct from a mechanistic repair that would alter the model's underlying logic. To attempt a true repair, a developer must often engage in fine-tuning or full retraining. Due to the opacity of the model, it is epistemically difficult to ascertain whether this process has corrected the underlying flaw, merely hidden it, or inadvertently introduced new and unknown failure modes [7], [8]. This contrasts sharply with the causal and localized nature of traditional software debugging.

The insufficiency of these technical methods should therefore be understood as a spectrum, because the legal determination of a reasonable standard of care is inherently risk-stratified. For some lower-stakes applications where harms are bounded, a documented heuristic audit might provide sufficient evidence of diligence. This calculus must also account for the effects of scale. A system not classified as high-risk in terms of catastrophic potential may still pose a significant societal risk when deployed to millions of users, where even low-probability edge cases will manifest and create substantial integrated harm.

Ultimately, for the highest tier of risks involving catastrophic or systemic consequences, a portfolio of technical methods alone is insufficient to provide the necessary proactive assurance. This tier includes systems that control critical infrastructure, applications with biosecurity implications, and the pursuit of artificial general intelligence. In these contexts, the potential for novel or adversarially concealed functions means that no combination of retrospective or statistical methods can provide the required proactive guarantee. It is for this tier of risk that the limitations of heuristic auditing become insurmountable barriers.

The practical limitation is one of incomplete coverage. While foundational heuristic methods like causal scrubbing and automated circuit discovery (ACDC) were demonstrated on smaller models [103], [104], subsequent work has begun to extend these techniques to transformer models with billions of parameters by leveraging strong task-specific and structural assumptions to prune and prioritize candidate subcircuits, reducing the search space to a tractable size [105], [106]. However, recent peer-reviewed benchmarks show that even medium-scale LLMs struggle to recover pre-defined circuits, which are subgraphs manually annotated by researchers and used to evaluate structural reasoning [107]. Despite targeted prompting and structured evaluation, current models achieve only limited success on these tasks [107].

Crucially, this limited success does not imply that all internal mechanisms can be reverse-engineered [107]. It reflects only the fraction of circuits the auditor already knew to look for [107]. Any emergent, novel, or adversarially hidden pathways, whether they exist in the remaining percentage or fall entirely outside the pre-specified search space, would escape these methods. As a result, while heuristic auditing is a valuable stepping stone, it offers no guarantee

of comprehensive coverage. The fundamental verification gap in frontier-scale models remains wide open. This leaves enterprises in a difficult position, they can use heuristic tools for partial checks, but must operate with the knowledge that unknown, potentially critical flaws could exist that their internal validation processes are fundamentally incapable of discovering.

The utility of heuristic methods in a high-stakes regulatory context is further limited by a fundamental epistemic problem which is the absence of a reliable method for validating the heuristics themselves. Mature engineering disciplines gain confidence in heuristics through measurement against extensive historical data and a stable physical reality [101]. The effectiveness of an AI safety heuristic, however, cannot be quantified with similar confidence [18]. The very novelty and emergent nature of the risks being audited, such as deceptive alignment, means there is no independent ground truth against which to calibrate these tools [8]. This creates a validator's dilemma, where any claims of high confidence are inherently speculative and cannot form a robust basis for legal or regulatory assurance.

This challenge is compounded by the adversarial nature of high-stakes deployment. The moment a specific heuristic becomes a standard for regulatory compliance or liability protection, it ceases to be a reliable test and instead becomes a target for adversarial gaming [67]. A developer could, intentionally or unintentionally through optimization, train a model to specifically pass the checks of a heuristic audit while failing catastrophically on inputs outside that narrow distribution [18]. This dynamic suggests that a reliance on heuristic testing could paradoxically degrade safety by creating a false sense of security and providing a roadmap for concealing the most dangerous capabilities, a phenomenon demonstrated in research on the persistence of backdoors through safety training [8].

A governance strategy that relies exclusively on heuristic auditing, however, is insufficient for addressing the problem of emergence. Heuristics are fundamentally retrospective tools designed to find known or anticipated flaws. While they are an indispensable component of any defense-in-depth approach for identifying common errors and improving general robustness, their efficacy diminishes when confronted with novel risks. The defining challenge of frontier AI is the unpredictable emergence of new capabilities and failure modes that were not programmed or foreseen. A heuristic cannot find a risk its designers have not yet conceived. Any governance strategy that depends solely on such tools will therefore remain reactive, capable of identifying known types of flaws but unable to provide proactive assurance against unknown unknowns.

3.4 A Comparative Analysis of Assurance Paradigms

An objection may be raised that other safety-critical fields manage extreme risks without

requiring complete formal verification. Disciplines like aerospace engineering and pharmaceutical development utilize a mature portfolio of extensive testing, rigorous design protocols, and statistical quality controls to achieve high levels of safety. While this approach has been effective for conventional systems, a granular comparison reveals that advanced AI presents a qualitatively different set of assurance challenges that limit the sufficiency of these established paradigms when used in isolation.

Table 1: A Comparative Analysis of Assurance Challenges in Safety-Critical Fields

Assurance Challenge	Aerospace Engineering	Pharmaceutical Development	Advanced AI Systems
Nature of System	Complex but governed by stable, empirically verifiable principles of physics.	Complex, but governed by principles of biology and chemistry that are stable and characterizable.	Adaptive and emergent. Internal logic and capabilities can change in response to data and interaction.
Operational Domain	Designed and tested for a well-defined and bounded operational envelope (e.g., flight conditions).	Tested for a specific set of clinical indications and patient populations.	Open-ended and effectively infinite input space. Deployed in dynamic, unpredictable environments.
Nature of Failure	Failure modes can often be traced to known causal chains (e.g., material fatigue, design flaw).	Adverse effects are stable properties that can be discovered through statistical methods in clinical trials.	Potential for novel, emergent failure modes not present in training and not traceable to discrete design flaws.
Potential for Deception	The system is passive. It does not strategically conceal its flaws during testing.	The system is passive. It does not strategically alter its chemical properties in response to testing.	The system can be adaptive and potentially strategic. It can learn to conceal dangerous capabilities during evaluation (e.g., sleeper agents).

These qualitative differences are not simply theoretical. They are already shaping the practices of expert agencies that regulate safety-critical systems. The U.S. Food and Drug Administration (FDA), for instance, has formally recognized that AI and machine learning software requires a novel regulatory paradigm distinct from that used for traditional software [108]. The agency's approach is based on the understanding that an AI model's performance can degrade in response to shifts in real-world data distributions, a challenge that requires continuous oversight [109]. In response, the FDA has developed a “total product lifecycle” framework that emphasizes post-market monitoring [109]. This has led the agency to create entirely new regulatory mechanisms. One such mechanism is the “Predetermined Change Control Plan” (PCCP), which prospectively authorizes certain future model modifications to account for data drift and represents a concept that does not exist for conventional software [108]. This precedent provides strong, real-world evidence that AI's adaptive and emergent nature necessitates a new governance approach.

As the comparison illustrates, the core distinction is not one of complexity, but of adaptability and inscrutability. The properties of a pharmaceutical compound and the performance of an aircraft are governed by stable physical laws. While these properties may be difficult to fully characterize, they do not strategically alter their behavior in response to the testing process itself. An advanced AI system, in contrast, can adapt based on its environment, and its open-ended operational domain means that pre-deployment testing can never cover the vast space of potential real-world inputs [7].

Most importantly, the potential for an AI to actively conceal misaligned goals or dangerous capabilities during evaluation represents a qualitative shift in the nature of the verification problem [8]. Conventional engineering paradigms are designed to characterize the properties of a complex but passive system. They are not designed to provide assurance for a system that can learn to adapt its behavior in response to the evaluation process itself. This unique combination of factors means that even a robust portfolio of traditional assurance methods leaves a significant residual risk. It necessitates a complementary governance approach that does not depend exclusively on the sufficiency of these established technical paradigms.

3.5 Confounding Network Dynamics

The theoretical challenges of circuit discovery are compounded by observable, emergent network phenomena. One such phenomenon is the hydra effect, where ablating one component of a model causes other components to compensate, preserving a capability and masking the ablated part's true function [105]. For a business using AI for a critical internal process, this means that even if a flaw is identified and seemingly fixed, the model could self-repair in a way that preserves the flawed behavior, creating a false sense of security for the internal audit team. This

dynamic creates a significant supply chain risk for an enterprise procuring a third-party model. The vendor may, in good faith, patch a reported vulnerability. Yet, the hydra effect could cause the model to preserve the unsafe capability through a different computational pathway, rendering the vendor's patch ineffective and leaving the customer unknowingly exposed. This is related to functional redundancy, where multiple network components can perform similar tasks or encode overlapping predictive functions [110], [111]. Another challenge is representation drift, where a model's internal representations of concepts shift based on conversational history [112], [113].

These confounding dynamics reveal a conceptual flaw in applying the traditional software bug fix paradigm to AI governance. When a heuristic audit discovers a flaw, it is not identifying a discrete line of code that can be patched. It is identifying an emergent property of a complex, holistic system [9]. Attempting to remove this flaw, for instance through fine-tuning, can trigger compensatory dynamics like the hydra effect, where the unsafe capability is preserved through different computational pathways [105]. The only potential remedy is often complete retraining, a prohibitively expensive process that may simply trade a known flaw for a new, unknown one [7], [114]. This breaks the iterative test-and-patch cycle that underpins traditional software assurance, making pre-deployment process governance a more tractable and critical focus.

Furthermore, the computational paths within these models are not always static. The sensitivity of attention mechanisms is a confounding factor, and while new methods show promise in tracing information routing through their non-linear interactions, the task remains exceedingly difficult for most circuit discovery tools [115]. This is complicated by innovations like input-dependent prompting, where a trainable module dynamically alters the computational path for each unique input, creating a moving target for auditors [116], [117]. Finally, there is growing evidence that models can develop implicit discrete state representations to perform complex, sequential reasoning without verbalizing intermediate steps [118]. The emergence of these internal, symbolic-like states challenges auditing approaches that assume linear feature representations [118]. These dynamics create a moving target for auditors attempting to create a static map of a model's internal mechanisms. As models scale, they exhibit emergent abilities, which are capabilities that are absent in smaller models and can appear unpredictably [19], though the nature of this emergence is debated [119]. This scaling also causes a combinatorial explosion in the number of potential computational pathways, creating an intractable search space for auditors seeking to map the model's mechanisms [16], [120]. Governance strategies should therefore be designed with the assumption of persistent opacity.

Table 2: Key Challenges in Mechanistic Interpretability

Challenge	Definition	Implication for Verification/Auditing

Polysemanticity & Superposition	The phenomenon where a single neuron activates for multiple unrelated concepts (polysemanticity), often because the model represents more features than it has neurons (superposition) [77], [78].	Makes it impossible to assign a single, clear meaning to a neuron's activation, confounding efforts to reverse-engineer a model's logic by examining individual components in isolation.
Circuit Intractability	The formal proof that finding a minimal, globally correct computational subgraph (a circuit) that explains a model's behavior is computationally intractable (Σ_2^p-complete) [16].	Establishes a fundamental, mathematical barrier to complete verification. Auditors can find counterexamples to a claimed explanation but can never prove its universal correctness.
Hydra Effect (Self-Repair)	The emergent capability of a network to compensate for the ablation (removal) of one component by dynamically re-routing computations through other components, preserving overall function [105].	Masks the true function and importance of individual components. An auditor might incorrectly conclude a component is non-critical because its removal does not immediately degrade performance, leading to a premature sign-off on a safety audit.
Representation Drift	The phenomenon where a model's internal representations of concepts are not static and can shift based on factors like conversational history or context [112], [113].	Undermines static analysis. A map of a model's internal concepts generated at one point in time may not be valid later, making long-term assurance difficult.
Input-Dependent Computation	Advanced architectures where the computational path	Creates a moving target for auditors. There is no single, static

	through the network is not fixed but is dynamically altered for each unique input [116], [121].	circuit to analyze, since the circuit itself changes with every input, exponentially increasing the auditing challenge.
Implicit Discrete States	The ability of models to develop internal, symbolic-like states to perform sequential reasoning without verbalizing intermediate steps [118].	Challenges auditing tools that assume linear or continuous feature representations. These hidden thought processes are invisible to standard interpretability methods.

4. The Regulator's Dilemma: Crafting Rules for Intractable Models

The technical limitations described in the preceding sections do not imply that technical auditing is without value. On the contrary, heuristic and behavioral tests are indispensable for identifying known flaws and improving the general robustness of AI systems [56]. The central argument of this paper, however, is that these technical methods are necessary but not sufficient for high-stakes applications. The verification gap creates a residual risk of novel or adversarially concealed failures that technical tools alone cannot eliminate. The process-based governance framework detailed in this section should therefore be understood as an essential complement to technical auditing. It provides a tractable basis for accountability that addresses the risks that persist even after technical methods have been exhausted.

The emerging and computationally challenging state of AI assessment technology creates a dilemma for any entity tasked with its governance [13]. Along with this technical challenge there is a political and institutional one, which is whether to govern AI through a new, specialized agency or by adapting the mandates of existing bodies [122], [123]. However, this paper argues that regardless of which institutional path is ultimately chosen, any regulator will confront the same core problems. The verification gap, and the technical uncertainty it creates, makes traditional product-based oversight difficult and fosters the ideal conditions for compliance theater, where the performance of auditable procedures becomes a convenient substitute for the intractable work of substantive risk mitigation, a practice now being scrutinized by regulators as AI audit washing [66], [67], [124], [125]. For the enterprise, this same uncertainty creates an internal governance dilemma, forcing risk and compliance functions to oversee a technology that they cannot independently validate [2]. This section explores how this reality constrains the

design of any effective regulatory framework.

4.1 From Product Safety to Process Governance

The framework proposed in this paper rests on a core assumption that requires justification. The assumption is that auditing a human-led development process, while challenging, is a more tractable problem than auditing the final AI model. This paper's argument rests on a direct comparison of the available assurance methodologies. The first, formal verification of the final model, is a known impossibility due to computational intractability [16]. The primary technical alternative is a portfolio of heuristic methods. These are best characterized as a phenomenological approach that empirically tests a model's input-output behavior to find flaws [53]. While this is an essential and valuable part of any safety workflow, whether it can provide a legally sufficient standard of 'reasonable' safety for high-stakes or large-scale systems is a fundamentally open and unresolved question. We currently lack the empirical evidence to know. This uncertainty is itself a direct consequence of intractability. We cannot be certain our heuristic tests are comprehensive, precisely because the underlying space of potential failure modes is too vast to map. Faced with one impossible method and one fundamentally uncertain method, a prudent governance framework must be supplemented by a third option. That option is the audit of the human-led process, an established methodology for assigning accountability for decisions made under uncertainty [66], [126]. A focus on process governance is therefore a rational and necessary component of any framework, shifting a portion of the accountability problem from an intractable or uncertain technical domain to a difficult but manageable forensic one.

The defense of process-based auditing rests not on the incentives for deception, but on the qualitative nature of the evidentiary trail. A substantive audit of an AI development lifecycle is a forensic investigation into a vast and deeply interconnected dataset. This evidentiary trail includes version control systems with cryptographically signed and timestamped commits, records of peer code reviews, project management tickets detailing bug fixes and feature development, logs from automated testing and deployment pipelines, and internal communications [127]. The challenge for a fraudulent actor is not merely to fabricate a single artifact, but to ensure forensic consistency across this entire high-dimensional system of records. Fabricating a coherent history that aligns across all of these sources presents a significant forensic challenge [128].

A further challenge, however, is ensuring the authenticity of this evidentiary trail against a sophisticated adversary who controls the information environment. To address this, a regulatory framework should not rely on trust, but on verification, using a methodology that is both robust for regulators and fair to developers. This paper proposes a multi-pronged methodology as a starting point. First, a technical root of trust could be established by requiring that critical

logging systems in the development pipeline operate within secure hardware enclaves like Trusted Execution Environments (TEEs) [129], [130]. This would produce a cryptographically verifiable log that not only allows an auditor to confirm the integrity of the evidence, but also provides honest developers with an unimpeachable record to defend against unfounded accusations of tampering [129], [130]. This technical assurance, however, is only a foundational layer. A sophisticated adversary could seek to compromise the entire evidence-generating environment [131]. The resilience of the audit process should be derived from a multi-layered methodology designed to function under adversarial assumptions. This approach draws from principles developed for High-Reliability Organizations (HROs), which maintain safety in complex, high-stakes environments through a preoccupation with potential failures [132]. In this context, the audit itself must be subject to a form of second-order verification. This involves procedural cross-validation, such as structured interviews with development and testing personnel, designed not only to gather information but also to identify organizational pathologies that are often precursors to systemic compliance failures [128]. Auditors would be trained to detect inconsistencies between the verifiable log data and the human-reported development narrative.

The ultimate defense is not the theoretical impossibility of deception, but the creation of powerful economic deterrents that render systemic fraud an irrational business strategy. An effective regulatory regime must couple the audit process with severe and certain consequences for discovered fraud. The willingness to undergo such a deep, multi-layered audit becomes the price for obtaining the legal certainty of a statutory safe harbor, as detailed in Section 5. Any evidence of fraudulent compliance discovered during an investigation would trigger not only the immediate and permanent revocation of this safe harbor but also the imposition of punitive financial penalties and referral for criminal investigation. This system of robust verification and severe deterrence raises the cost of deception while protecting developers who engage in good faith.

Attempts to apply conventional product-safety frameworks to AI reveal foundational challenges [41]. Regulatory models based on investigating discrete incidents are difficult to apply when AI can cause harms that are subtle, cumulative, and widely distributed [50], [51], [133]. Furthermore, regulatory approaches that rely on pre-market approval are fundamentally undermined by the verification gap [134], [135]. As this paper argues, the formal intractability of auditing makes it exceedingly difficult for a regulator to gain sufficient assurance of a model's internal safety before it is deployed. This challenge is compounded by the adaptive nature of these systems, which means that a model's behavior can change after its release [75], [136]. When the final product is an opaque and intractably complex system, a viable path for regulation involves fundamentally changing the approach to oversight. This situation necessitates a shift in regulatory focus from exclusively evaluating the final product to scrutinizing the process that

created it [57], [137].

This move is conceptually similar to the established 'Quality by Design' (QbD) paradigm, a mature regulatory science principle that emphasizes proactive process control over reactive product testing [137], [138], [139], [140]. While the principle is not new, its application to AI development requires a significant adaptation because the AI production process is qualitatively different from the deterministic physical manufacturing for which QbD was designed. At least four distinctions are critical. First, AI development is an inherently stochastic process where identical training pipelines can produce models with different emergent behaviors [7], [19]. Second, the quality of the final product is inextricably dependent on the provenance and integrity of vast datasets, making data a unique raw material with its own complex supply chain [30], [141]. Third, the process creates a system whose capabilities can emerge unpredictably and exceed its initial specifications, a challenge with no direct analogue in traditional manufacturing [19]. Finally, the output is not a physical object but an intractably complex digital artifact whose internal properties cannot be fully inspected [9], [16]. Regulating the process that generates the AI model is therefore an essential tool for accountability when the internal properties of the model cannot be directly and reliably measured.

A framework for AI Process Control could mandate the tracking of specific metrics at each stage. During the data ingestion and pre-processing phase, metrics could focus on data quality and potential bias. This would include tracking statistical measures of data drift over time such as the Population Stability Index or Wasserstein distance to detect shifts in the input data distribution before they degrade model performance [75], [142]. It could also involve fairness metrics that measure the difference in population size or the proportion of positive labels across sensitive demographic groups to identify potential sources of bias in the training data itself [143].

During the model training and validation phase, metrics would track model performance and robustness. This includes standard measures like accuracy, precision, recall, and F1 scores on holdout datasets [144]. It could also include tracking the rate of bias detection, which measures the percentage of biased outcomes that are identified and mitigated during testing [145], [146].

Finally, in the post-deployment and monitoring phase, metrics could shift to operational and user-centric KPIs. These include technical measures like system uptime model latency and error rates [56]. They could also incorporate direct feedback from end users and from others affected by the model's outputs [56].

The selection and reporting of these metrics should not be static. An effective governance system would establish a meta-process guided by frameworks like the NIST AI RMF's Measure and Manage functions for continuously evaluating which metrics are most salient for a given application and risk context [56]. This would create an adaptive system of oversight where the

methods of verification evolve alongside the technology itself [56]. This feedback loop is critical in ensuring a quality by design system is measuring and assessing the correct metrics, and not simply becoming another act in the compliance theater where checking boxes substitutes for advancing safety [66], [67].

4.2 An Auditable Framework and Its Enforcement

Process-based governance is embodied in frameworks like the NIST AI Risk Management Framework (AI RMF) [56] and the international standard ISO/IEC 42001 [147]. These frameworks encourage the evolution of internal best practices into a credible, market-facing signal of trustworthiness that serves both external and internal functions [56], [147]. For regulators and customers, they provide a verifiable account of due diligence. For the organization itself, they create the internal discipline needed to manage the risks of deploying a technology that is not fully understood [56], [147]. These frameworks also provide a structure for implementing meaningful human oversight, moving beyond the procedural formality that can arise when reviewers lack insight into a model's function [56].

This is analogous to trusted standards in other mature industries, such as the UL listing for electrical safety or the IIHS Top Safety Pick award for automobiles [148], [149]. While the complexity of AI auditing is greater, the underlying market principle is the same. These third-party certifications are not merely compliance exercises, but they are also competitive differentiators that companies proactively seek and advertise to build consumer trust and unlock market access [150]. In a similar way, a verifiable certification against a standard like ISO/IEC 42001 can serve as a powerful signal of process integrity for AI systems, providing a mechanism for developers to demonstrate their commitment to safety and for enterprise adopters to manage both external liability and internal operational risk [150]. For an enterprise, such a certification can serve as a crucial tool for supply chain risk management [151]. It provides a degree of assurance that an upstream model developer has followed a rigorous, auditable process, mitigating some of the risks associated with un-auditable data provenance and opaque design [151].

For such a signal to be meaningful, however, its integrity must be protected. A purely voluntary framework risks dilution, where inconsistent application or superficial check-the-box audits can devalue the certification for everyone [67]. To ensure that a certification like ISO/IEC 42001 remains a credible and trusted standard, the system must be coupled with mechanisms that ensure consistency and rigor. This includes robust third-party audits requiring auditable documentation of risk assessments and lifecycle management [147], [151]. Crucially, certification should be linked to a suite of tangible benefits to create a powerful market incentive

for genuine compliance.

Such benefits could be modeled on incentives from other regulated domains. The most direct analogy is federal procurement, where compliance with cybersecurity standards is already a mandatory prerequisite for government contractors handling sensitive information [152], [153]. For example, the Defense Federal Acquisition Regulation Supplement (DFARS) mandates that contractors implement the security requirements of NIST SP 800-171 to protect Controlled Unclassified Information [152]. Making ISO/IEC 42001 certification a requirement for any AI vendor working with the government could provide a powerful incentive for verifiable process integrity. For smaller businesses, a tiered system of regulatory fees, where certified companies pay less, could be adopted, mirroring the FDA's approach to medical device user fees [154]. Finally, such certification could serve as a powerful evidentiary shield against claims for punitive damages [155], [156], [157], [158], [159]. By demonstrating adherence to a rigorous, recognized standard of care, a company can rebut allegations of malice or reckless disregard that are typically the prerequisites for punitive awards [160, Sec. 908(2)], [161]. As a practical consequence, insurers view that same evidence of industry-benchmark compliance as a risk‑mitigation factor and often extend more favorable premium rates or deductible structures to certified entities [162, pp. 213–215].

An effective framework would mandate specific artifacts like Datasheets for Datasets [141] and Model Cards [163] transforming the abstract process into a verifiable paper trail. A key component of such a framework is the institutional body responsible for oversight, a challenge being addressed differently by various nations. The United Kingdom, for example, has proposed the creation of a centralized AI Authority through its Artificial Intelligence (Regulation) Bill [164]. This body would be tasked with ensuring alignment among existing regulators, conducting gap analyses, and facilitating sandboxes, offering a middle way between the EU's comprehensive legislation and US fragmentation [164].

A framework of earned safe harbors, however, creates an inherent tension. The powerful incentive to obtain a certification can also become an incentive to achieve it through superficial compliance [165]. To counter this risk and protect the integrity of the standard, the system must be coupled with a credible and robust enforcement methodology. This methodology must be designed to make fraudulent or grossly negligent compliance an economically irrational business strategy.

A starting point for such an enforcement regime would combine intelligent re-audits with severe consequences for fraud. First, an adaptive re-audit system would supplement random, unannounced inspections with deep forensic investigations triggered by specific risk signals [166], [167], [168]. These could include substantive signals, such as a high rate of real-world safety incidents, anomalous signals, like a significant divergence between a model’s tested and

deployed performance, and human intelligence, such as credible whistleblower reports [168]. Second, the consequences for discovered fraud must be sufficiently severe to provide a powerful deterrent [169]. A regulatory framework could include mechanisms such as the immediate and permanent revocation of a company's certification, referral to law enforcement for criminal investigation, and the imposition of punitive financial penalties, such as treble damages for any civil harms that occurred during the period of fraudulent compliance [170], [171], [172].

4.3 Case Study: Protecting the Market with Existing Authority

To illustrate how the integrity of a process-based framework could be protected within an existing legal structure, one can examine the potential authority of an agency like the Federal Trade Commission (FTC) [173], [174], [175]. This analysis is not a prescriptive claim that the FTC is the optimal regulator, but rather a case study demonstrating that the tools to protect the market from deceptive claims and ensure the credibility of process standards may already exist.

The FTC could use its authority under Section 5 of the FTC Act, which prohibits unfair or deceptive practices, to argue that making unsubstantiated safety or trustworthiness claims without a robust, RMF-aligned audit trail constitutes a deceptive practice [176], [177], [178]. This approach would be grounded in the FTC’s long-standing Unfairness Doctrine. Under this framework, a practice is deemed unfair if it causes substantial consumer injury, is not outweighed by countervailing benefits, and is not reasonably avoidable by consumers themselves [179]. Harm from an opaque AI system fits these criteria because the injury can be significant, the lack of transparency offers no countervailing consumer benefit, and the harm is not avoidable by a user who has no means of inspecting the model's internal logic [56], [143], [176], [177], [180], [181]. This logic can be extended to protect enterprise customers who are sold AI systems based on misleading claims of safety, only to discover that the system introduces significant, un-auditable risks into their own operations. The long-term viability of such an enforcement strategy would also be shaped by the judiciary's willingness to defer to the agency's interpretation of its statutory authority, a legal landscape recently transformed by the Supreme Court’s decision in *Loper Bright Enterprises v. Raimondo* [182].

The agency has already established a clear precedent for this type of market protection. The FTC's April 2025 consent order against accessiBe for deceptive claims about its AI-powered web accessibility tool demonstrates its willingness to act against AI-washing [183]. While relying on existing authority is not without challenges, such as potential industry pushback and questions of agency bandwidth, the concept is plausible [184], [185]. Inter-agency coordination, such as a memorandum of understanding between the FTC and NIST, could bolster this role by linking NIST's technical expertise to the FTC's legal authority [186].

This creates a virtuous cycle where a clear technical standard enables the policing of deceptive claims, and the enforcement against such claims protects the value of the standard for the entire market [126], [168]. This example demonstrates that the mechanisms to ensure a trustworthy AI ecosystem are not merely a theoretical construct, but a plausible path forward within the current U.S. legal system [176], [183].

5. The Unsettled Landscape of Legal Liability

The consequences of the verification gap extend far beyond the technical challenges of safety and alignment [41]. As a fundamental property of current AI systems, it introduces a level of structural uncertainty that impacts the design of regulatory frameworks and could shape strategic decisions of businesses and consumers in how and when to use AI [2], [187]. Companies that develop or deploy these systems must account for a new category of operational and reputational risk stemming from a technology whose behavior cannot be fully guaranteed [188]. This creates a new set of considerations for product development, deployment, insurance, and corporate governance [188], [189], [190], [191], [192]. While these business and strategic implications are substantial, this section will focus on one of the most immediate and disruptive consequences of the verification gap, its impact on the legal system's established frameworks for assigning responsibility for harm.

5.1 The Inadequacy of Traditional Tort Law

The primary mechanism for redressing harm in the United States is the tort system, which relies on the doctrines of negligence and product liability [88]§ 39. Both face challenges when applied to AI harms [41]. The central issue is the direct connection between the technical verification gap and a legal evidentiary gap, and the opacity of AI systems, which can make it extraordinarily difficult for a plaintiff to prove fault and causation [42].

Under a negligence standard, a plaintiff must show a developer failed to exercise reasonable care in protecting others against foreseeable risks [42], [193, Sec. 282. Negligence Defined]. In areas where no reliable technical test can verify a model’s internal safety, adherence to an auditable development process can become the industry-accepted benchmark for judging care [57]. However, care must be used to ensure that this dynamic does not fuel compliance theater in which companies can check off a standardized process rather than prove actual safety [66], [126], [194].

It should be noted that the incentive to gain a legal shield against liability by demonstrating adherence to a regulatory process is a well-established principle in U.S. tort law. Federal vaccine

law provides an example of this by shielding manufacturers from liability for unavoidable adverse effects when they comply in all material respects with the Food Drug and Cosmetic Act and its regulations [157]. Likewise, the Supreme Court in Riegel v. Medtronic held that medical device makers who follow FDA premarket approval standards may preempt state tort claims [195], [196]. A company can therefore argue that following a process like the NIST AI Risk Management Framework fulfills its duty of care, effectively substituting procedural diligence for verifiable safety [56], [197, Sec. 3. Negligence, 13. Custom]. This dynamic exposes the central design challenge for any process-based regime. The very regulatory models that provide a template for tractable oversight can, if the corresponding liability rules are poorly structured, also weaken the incentives required to ensure that oversight is substantive. The critical legal question then becomes whether this compliance was a genuine effort to mitigate foreseeable harm, or merely a procedural shield that obscures a failure to grapple with the model's actual risks [42], [197, Sec. 13 cmt. c, 16(b). Statutory Compliance].

Product liability law, which can impose strict liability for defective products, faces persistent challenges when applied to software [198]. A central unresolved issue is whether standalone software qualifies as a product under the Restatement (Third) of Torts, which traditionally limits product liability to tangible personal property [199, Sec. 19. Definition of "Product"]. Courts in the United States remain divided on this question, with doctrinal inconsistencies across jurisdictions and no clear consensus [198], [200], [201], [202], [203]. Recent scholarship has emphasized that AI systems and software applications disrupt traditional product liability frameworks by challenging assumptions about defectiveness, foreseeability, and control [204], [205]. Some courts have allowed product liability claims involving AI applications to proceed past the pleading stage, focusing on allegations of design defects rather than any settled classification of software as a product [204]. This judicial posture reflects a growing willingness to entertain liability theories where harm arises from system architecture rather than expressive output, but does not yet signal a doctrinal shift [198], [206]. Many jurisdictions continue to apply a risk utility test to determine if a design defect exists, which often requires plaintiffs to propose a reasonable alternative design and thereby reintroduces negligence-like elements into what is nominally a strict liability framework [199, Sec. 2. Categories of Product Defect], [205].

A critical unresolved question is how these doctrines apply when AI is used for an internal function, but its output contributes to an external harm [42]. For instance, if AI is used to help design a bridge or write code for a car's braking system, its unverified and flawed output could be integrated into a final product that later fails and causes injury. This creates a novel legal challenge. A plaintiff could argue that the defect or negligence originated in the unverifiable internal development process, even if the final product passed all standard human-led quality assurance tests. This scenario also challenges the legal sufficiency of human oversight. If the human reviewer is subject to automation bias or cognitive offloading and cannot inspect the AI's

internal logic, their review may be deemed a procedural formality rather than a substantive check, potentially re-exposing the enterprise to liability [207]. The use of an unverified AI in the supply chain introduces a new potential point of failure that complicates traditional analyses of foreseeability and reasonable care [42].

A potential turn toward heuristic auditing as a standard of care would not resolve these legal challenges, and could instead create a liability black hole that renders fault attribution impossible. In litigation following an AI-caused harm, a developer could argue its duty of care was met by using a certified third-party audit tool [197, Sec. 13. Custom]. The legal dispute would then shift to an intractable battle of the experts over the technical sufficiency of the heuristic itself, a question courts are ill-equipped to resolve [206], [208]. This deep technical uncertainty is unworkable for courts and insurers, who require a predictable basis for assigning risk and responsibility [209]. A legal framework focused on auditable human-led processes and clear liability defaults, as later proposed, is therefore essential to create the market certainty necessary for both innovation and accountability.

These doctrinal challenges are compounded by the practical realities of litigation. Even when legal rules are clarified through statutes or case law, the inherent complexity of AI technology poses a significant institutional hurdle [206]. The legal system has long struggled to resolve disputes involving advanced technologies, often resulting in a battle of experts [208], [210], [211]. In such cases, non-technical judges must navigate rulings at the intersection of law and technology, while juries must evaluate highly technical evidence and conflicting expert testimony to make factual determinations [212]. The recent history of U.S. patent law provides a useful illustration of this dynamic [213]. Over the past two decades, a series of Supreme Court decisions on patents has dramatically reshaped key aspects of the field [213]. These rulings, driven by judicial interpretation rather than legislative action, altered expectations within the patent community, which had ripple effects in the business world [214], [215], [213], [216]. A similar evolution could occur in the AI field. Regardless of the liability frameworks adopted, it is ultimately the judiciary that will determine their practical application. Therefore, any proposed solution must account for the real world institutional challenges inherent in interpreting and enforcing these frameworks.

5.2 The Search for New Liability Frameworks

5.2.1 The European Context and Its Lessons

The legal situation in the European Union has shifted dramatically with the adoption of the revised Product Liability Directive, which explicitly expands the definition of product to include software and AI systems, subjecting them to a strict liability regime [217]. This directive was

intended to be complemented by an AI Liability Directive (AILD), which would have further eased the path for claimants by creating a rebuttable presumption of causality [98]. However, the AILD was withdrawn in February 2025 [218], [219], [220]. Supporters in Parliament attributed the withdrawal to pressure from industry lobbyists who viewed liability rules as a threat to innovation while the European Commission cited a lack of agreement among member states and legislative complexity as the official reason for withdrawal [220], [219], [218]. The result is a fragmented legal landscape governed by 27 national tort regimes [221], [222], [223], [217].

The failure of the AILD highlights that the path to a stable liability regime is shaped by both economic interests and national regulatory philosophies. The European Union's governance model is often characterized by its reliance on the precautionary principle, which favors proactive regulation to prevent potential harms [224], [225]. The United States has historically favored a more reactive regulatory posture, often allowing innovation to proceed until concrete harms trigger a legislative or judicial response [225]. This reactive tendency, while innovation-friendly in the short term, creates the risk of a disproportionate policy overcorrection following a major AI-caused disaster. The history of the American nuclear power industry, which saw its growth stagnate for decades under the costly regulatory regime enacted after the Three Mile Island incident, serves as a cautionary example [226].

Within such a reactive system, the primary driver of change is often not proactive policy, but overwhelming economic pressure. The verification gap imposes a significant uncertainty tax on the economy, as the unquantifiable risks of deploying opaque AI systems chill investment and slow broad-based adoption by mainstream firms [2], [227], [228]. While this acts as a persistent drag, the trigger for a political resolution will likely be the market for risk transfer. Following initial high-profile liability events, insurers will be forced to price this uncertainty, leading to dramatically higher premiums or the introduction of broad exclusions for AI-related harms [229]. This market reaction will make the use of unauditable AI economically untenable for many firms.

This creates a powerful incentive for insurers and large enterprise adopters to act as de facto regulators, demanding auditable process standards as a precondition for coverage and commercial engagement [230], [231]. The market itself will begin to demand the very mechanisms of accountability this paper proposes. This dynamic ultimately compels a political response, because developers will eventually seek a legislative framework that offers a conditional safe harbor, which will be the only path to predictable insurability and broad market access. A pragmatic strategy for a reactive culture, therefore, is to engage in proactive regulatory learning through tools like the AI Liability Sandbox discussed in Section 7.2. By building institutional expertise before a crisis, policymakers can ensure the eventual, economically-driven regulatory response is measured and effective, rather than panicked and overbroad.

5.2.2 A Tiered Framework Informed by the Verification Gap

The analysis of the verification gap provides a diagnostic tool for adapting standard policy templates, like a tiered liability framework, to the specific challenges of AI. The central insight is that liability should be allocated based on a party's ability to mitigate the verification gap, an ability which resides almost exclusively in the proactive control over the development process. A framework designed around this principle would not treat all AI systems equally. Instead, it would align the standard of liability with the severity of the potential harm and the depth of the verification gap.

A crucial implementation challenge for this framework is the development of a clear rubric for distinguishing high-risk from general commercial applications. This paper does not propose a new rubric from scratch, as such classifications must evolve with real-world evidence. Instead, it proposes that the framework be operationalized by adapting the risk classification methodologies already developed in key governance frameworks. The EU AI Act, for example, provides a direct precedent for a risk-based system, designating certain use cases as high-risk, such as systems used in the management of critical infrastructure, in employment and hiring, or in law enforcement [38]. Similarly, the NIST AI RMF provides a detailed methodology that an organization would use to conduct its internal risk assessment, creating a structured analysis to present to a regulator who would make the final classification [56].

The core principle, reflected in these frameworks, is that risk classification must be determined by the context of use, not by the technology in the abstract [38], [56]. For example, a general-purpose code generation tool used by a developer for a non-critical application would be a general commercial use. The exact same tool, when integrated by a corporation into the software supply chain for an autonomous vehicle, would become part of a high-risk application. The entity that chooses to deploy the tool in a high-risk context therefore assumes the corresponding higher duty of care. This contextual approach provides a practical and legally coherent methodology for resolving the line-drawing problem.

For high-risk applications, the default standard should be strict liability. In these cases, the potential for harm is severe, and the foundational and practical gaps in verification are at their widest. The developer, by choosing to build and deploy such a system, is the only party with the process control to attempt to mitigate these gaps from the start. Strict liability appropriately places the risk on the actor best positioned to manage it and creates a powerful economic incentive to invest in safety measures that are commensurate with the potential for harm [232], [233]. To provide a path for legal certainty, this could be complemented by a formal administrative safe harbor, analogous to the FDA's pre-market approval for medical devices

[234]. Under this model, a developer of a high-risk system could voluntarily submit their model and internal process documentation to a designated expert regulator for a rigorous, confidential review. In exchange for this deep transparency, which serves to close the asymmetric gap for the regulator, an approval would grant the developer a statutory shield, shifting any subsequent tort claims against them from strict liability to a standard of negligence.

The viability of such an administrative safe harbor depends on the existence of a designated expert regulator capable of conducting the necessary technical reviews. Creating such an entity is a formidable governance challenge in itself, requiring significant political will, funding, and a strategy to attract and retain scarce technical talent while resisting regulatory capture [165], [184]. These are not minor obstacles, but their scale is precisely what justifies an iterative, evidence-based path to implementation. As detailed in Section 7.2, an AI Liability Sandbox provides the necessary first step. It functions as a controlled environment where a nascent regulator can build institutional capacity, develop evaluation methodologies, and gather empirical data on AI failures. This process of proactive regulatory learning is the most pragmatic path to developing the competent oversight body that a mature liability framework will eventually require.

For general commercial applications, the default standard should be a rebuttable presumption of causality. This rule is a direct legal response to the asymmetric gap. It presumes the developer is causally responsible for a harm because their exclusive process control makes it impossible for a plaintiff to prove causation within the opaque model. This presumption is rebuttable, however. A developer could shift the standard back to traditional negligence by affirmatively demonstrating adherence to a rigorous, auditable development process, such as through ISO/IEC 42001 certification and the production of verifiable artifacts [147]. This mechanism creates a direct market incentive for developers to adopt transparent processes that reduce the information asymmetry. In this framework, certification would not serve as an absolute bar to liability, but as powerful prima facie evidence of a reasonable standard of care. The burden would then shift back to a plaintiff to demonstrate with specific evidence that the certification itself was obtained fraudulently or that the developer knowingly concealed a flaw that fell outside the scope of the audited process.

Finally, for non-commercial research and open-source development, a full statutory safe harbor is essential. From a verification gap perspective, the open-source model is unique because it inherently reduces the asymmetric gap by making the model's architecture, and often its development process, available for public scrutiny [235], [236]. Imposing liability on the creators of these foundational public goods would create a chilling effect and perversely disincentivize the very transparency that helps mitigate the verification gap [237], [238]. Liability should instead attach to the downstream commercial entities that choose to build a product using an open-source model, since they are the actors placing final products into the

stream of commerce and assuming the responsibility to conduct their own due diligence [199, Sec. 1. Liability of Commercial Seller or Distributor for Harm Caused by Defective Products].

This structure is designed to directly counter the risk of compliance theater identified earlier. Unlike a blanket statutory shield that might preempt claims based on regulatory approval, the safe harbors in this framework are conditional and must be earned. For general commercial applications, the ability to rebut a presumption of causality is predicated on a defendant's ability to produce substantive, verifiable evidence of a rigorous development process. A mere certificate of compliance would be insufficient without the underlying artifacts to support it. Similarly, the administrative safe harbor for high-risk systems is not an automatic grant but a bargained-for exchange that requires transparency with an expert regulator. This high bar for earning a less demanding liability standard turns the safe harbor mechanism into a powerful incentive for substantive diligence, rather than a reward for procedural box-checking.

5.2.3 Evidentiary Standards for the Regulatory Bargain

The safe harbors proposed in this framework are a bargained-for exchange of legal certainty for substantive diligence. The evidentiary standards for this bargain do not have to be created from scratch. They can be modeled on established legal principles and existing safe harbor frameworks, particularly those developed in tort law to govern punitive damages against regulated industries. This approach is supported by a substantial body of legal and economic scholarship.

Authoritative tort law scholars like David G. Owen have extensively analyzed the regulatory compliance defense, arguing that while compliance with a rigorous regulatory scheme should not bar claims for compensatory damages, it should provide a strong shield against punitive damages [155], [199, Sec. 4 cmt. e]. The logic is that a developer who makes a good-faith effort to comply with an expert agency's standards is not acting with the malice or reckless disregard that punitive damages are designed to punish [160, Sec. 908(2)], [161]. Law and economics scholarship provides a theoretical justification for this, viewing evidentiary standards as tools for allocating the risk of error [199, Sec. 4 cmt. e], [232]. A heightened standard like clear and convincing evidence is used when the social cost of a mistaken verdict is high, such as wrongly punishing a good-faith innovator and creating a chilling effect on the market [90].

Applying these established principles, the evidentiary standards for the proposed framework would be structured as follows. For general commercial applications, a developer could rebut the presumption of causality by presenting evidence of process adherence. A plaintiff could still seek compensatory damages under a traditional negligence standard. To be eligible for punitive damages, however, the plaintiff would then need to meet the heightened standard of proving by

clear and convincing evidence that the developer's compliance was fraudulent or concealed a known flaw [155], [199, Sec. 4 cmt. e].

For high-risk applications, a similar principle would apply. Once a developer earns the administrative safe harbor that shifts claims from strict liability to negligence, a plaintiff could still recover compensatory damages by proving negligence. To obtain punitive damages, however, the plaintiff would again need to prove by clear and convincing evidence that the developer committed fraud on the regulator or acted with malice [90], [160, Sec. 908(2)]. Meeting this high burden would likely require evidence uncovered during a deep forensic re-audit of the kind described in Section 4.2, such as proof of inconsistencies between cryptographically verified development logs and other records.

5.2.4 Market Responses to Liability Uncertainty

This legal and economic uncertainty has caused markets to react to the unquantifiable risks of the verification gap through adjustments in both oversight and coverage for AI risks [209], [229], [230]. Insurers are carving some AI risks out of legacy policies and offering affirmative AI coverages that condition protection on demonstrable governance [229]. Market and coverage counsel report a surge in AI specific exclusions in directors and officers, cyber, and professional liability forms, while carriers also roll out endorsements and standalone AI products that expressly insure deployments meeting specified governance or testing standards [229]. Also, in some cases, underwriting is shifting to governance signals rather than relying solely on historical loss experience [230]. Brokers and carriers may require documented model testing, post deployment monitoring, incident response plans, and third party audit evidence as underwriting inputs that materially affect placement and pricing [230]. When standard insurance is unavailable for new AI risks, firms are creating more complex solutions [209]. They often build layered insurance programs that use reinsurance, create their own insurance subsidiaries (captives), or negotiate custom-written policies to manage risks that traditional plans will not cover [209].

5.3 The Open-Source Liability Trilemma

The political landscape is complicated by a debate over open-source versus closed-source AI [239], [240], [241]. This debate creates a liability trilemma, revealing a tension among establishing a strong liability regime, fostering public-interest innovation, and maintaining a competitive market [239], [242], [243]. Applying a uniform liability standard to every entity that contributes to an AI model would have a chilling effect on the non-commercial and academic open-source ecosystem [237], [238], [244], [245]. This ecosystem provides significant public

benefits by democratizing access, enabling independent auditing, and providing a competitive check on dominant firms [235], [236].

The legal distinction between non-commercial and commercial actors rests on the policy principle that liability should follow the placement of a product into the stream of commerce [199, Sec. 1. Liability of Commercial Seller or Distributor for Harm Caused by Defective Products], [246], [247]. This principle is not unique to products liability. It is a consistent feature of American law, reflected in the differing duties applied to merchants and private sellers under contract law [248] and the distinct standards of care required of professional and amateur actors in tort law [249].

Under this framework, an academic researcher who releases a model primarily as a research artifact is not placing a product into the market [88]§ 100. A company that subsequently integrates that model into a commercial service, however, is doing precisely that and thereby assumes the attendant duties of a commercial producer [199, Sec. 1 cmt. g]. A targeted liability framework therefore does not create a blanket immunity for potentially harmful technology. It instead correctly assigns responsibility to the specific actor that commercializes the technology [250], [243, pp. 211–240]. The logic of this approach targets the commercial deployer, the entity best positioned to implement technical safeguards and to insure against the financial consequences of potential harms [87, pp. 941–973], [162, pp. 213–215], [233]. Accordingly, an effective safe harbor policy must be precise, shielding non-commercial research and development while holding commercial deployers to the high standard of care expected of any entity that introduces a powerful technology into the public sphere [251].

Table 3: A Comparative Analysis of AI Liability Frameworks

Key Provisions

- **Traditional Negligence:** Must prove duty of care, breach, causation, and damages. Standard of care is based on industry practice.
- **Traditional Product Liability:** Strict liability for defective products. Historically, standalone software often not considered a product.
- **EU Revised Product Liability Directive (PLD):** Strict liability (no-fault). Explicitly defines software and AI systems as products. Liability extends to post-deployment updates and lowers the burden of proof for claimants.
- **Proposed Tiered Framework:** Risk-stratified liability defaults, such as strict liability and a rebuttable presumption of causality, paired with conditional safe harbors that must be earned through adherence to rigorous and auditable process standards.

Applicability to AI

- **Traditional Negligence:** Applies to AI services and deployment.
- **Traditional Product Liability:** Uncertain, but a judicial shift is underway. May apply if AI is embedded in hardware or if courts continue to treat standalone software as a product.
- **EU Revised Product Liability Directive (PLD):** Directly applies to all AI systems placed on the EU market. It is now the primary EU-level instrument for AI harm.
- **Proposed Tiered Framework:** A proposed future framework designed specifically for the AI value chain.

Primary Challenges for Plaintiffs

- **Traditional Negligence:** Proving breach of a rapidly changing standard of care. Proving causation in a multi-actor supply chain with an opaque model. Proving that reliance on an unverified AI, or that the human oversight of it, was negligent.
- **Traditional Product Liability:** Proving AI is a product Proving a defect in a probabilistic system, often via a complex risk-utility test.
- **EU Revised Product Liability Directive (PLD):** Fewer challenges, but it excludes pure economic loss. Defectiveness and causation can be presumed in cases of technical complexity.
- **Proposed Tiered Framework:** Meeting a heightened evidentiary standard to overcome a defendant's evidence of process adherence or to pierce an earned administrative safe harbor.

6. The Economic and Political Context of AI Governance

The technical and legal challenges of AI oversight exist in a complex political and economic context shaped by intense global competition, powerful industry interests, and contentious debates over the relationship between regulation and innovation. These interests create significant opposition that can impede efforts to establish effective governance [252]. The core tensions are not just domestic but geopolitical, touching on issues of data sovereignty, AI supply chain concentration, and the enormous environmental and capital costs of scaling these systems [252], [253].

6.1 The Innovation-Regulation Nexus and the Cost of Uncertainty

The debate over AI regulation is often framed as a trade-off between safety and innovation [254], [255], [256], [257], [258], yet the current AI landscape presents a paradox where a major investment boom, with U.S. private AI investment reported at $109.1 billion in 2024, coincides

with substantial regulatory uncertainty [1]. Frontier AI developers are advancing capabilities across multiple domains, including technology and legal questions that arise from rapid deployment [259], [260], [261]. They are making massive capital expenditures on computing infrastructure [114], [262], [263], [264], [265] while navigating unsettled legal questions, including fair use issues concerning training data [260], [266], [261], [267]. These capital costs also create significant environmental externalities, as the massive energy and water consumption required for training and operating frontier models is becoming a significant policy concern for governments managing grid stability and climate goals [268]. A portion of this consumption is an implicit cost of the verification gap itself, because the absence of efficient auditing tools can force developers to rely on redundant, energy-intensive testing cycles to build confidence in a model's safety [7]. The opacity of these models means that unintended behaviors, including offensive or infringing outputs, can occur despite developers' mitigation efforts [11]. To mitigate such risks, developers commonly implement safety filters and behavioral guardrails [117], [269], [270]. However, this often incurs an alignment tax, meaning additional costs such as reduced performance, longer development time, and higher compute expenses that arise when aligning an AI system with human values and intentions [271], [272]. This can lead models to refuse reasonable requests or to produce overly cautious and less useful responses, converting a technical control challenge into a complex functional and legal problem at the frontier of innovation [117], [271].

This focus on frontier development overlooks a crucial distinction between the creators of foundation models and the much larger ecosystem of mainstream enterprise adopters. While the race to build more powerful models continues, regulatory uncertainty creates a barrier for end users such as hospitals, financial institutions, and many other companies that must deploy these tools in mission-critical or public facing operations [2], [228]. For this silent majority of the economy, the lack of clear and stable rules may be a primary deterrent that chills broad-based integration of AI needed for long term productivity gains [2], [123], [228], [273].

History offers a useful analogy. The mass adoption of the automobile was not unlocked solely by technical innovation, but by the gradual development of a stable regulatory framework including traffic laws, driver licensing, and clear rules for liability [274], [275], [276], [277], [278]. That framework gave users the certainty to operate vehicles safely and predictably, and in turn created the trustworthy mass market necessary for the industry to flourish. Similarly, for AI, a primary barrier to widespread enterprise adoption may be the lack of a clear legal and regulatory framework that provides certainty for downstream users of these systems [2], [228].

6.2 Potential Market Failures: Information Asymmetry and the Compliance Moat

The verification gap creates a condition of pervasive uncertainty for all market participants [1], [4], [15], [279]. While the intractability of complete verification means that even developers cannot possess full certainty about a model's behavior, developers do have exclusive access to their training data, design choices, and internal use and testing results [16], [14], [7], [145]. This dynamic generates a severe information asymmetry [280], [281]. Buyers of AI systems lack access to the proprietary know-how and internal data needed to independently assess a model's safety, quality, or the true merit of a vendor's compliance claims [14], [30], [280].

This information asymmetry has the potential to create the conditions for a market for lemons, a dynamic where low-quality goods can drive out high-quality ones [281]. When buyers cannot reliably distinguish safer, more robust models from less safe ones, they become unwilling to pay a premium for invisible safety features they cannot verify [281]. This disincentivizes developers from investing in costly, rigorous safety measures that do not produce a marketable feature, creating a market that pulls toward the lowest common denominator on safety [15], [281]. This outcome can occur even in the absence of bad intentions on either side, and since complete verification is not possible, relevant portions of the information asymmetry may persist since they directly implicate the company’s perceived competitive advantage [87], [165], [7], [281]. In such an environment, auditable compliance may serve as the public signal for quality to help solve this market failure, but this approach carries the inherent risk of devolving into compliance theater in which the audit signals diligence without guaranteeing it [66], [67], [282].

A separate but related market challenge arises from the high cost of generating credible quality signals [242], [281]. Whether due to government regulation, industry-led certifications, or even a push to solve the information asymmetry problem, such quality signals could require expensive testing, documentation, and third-party audits [283]. While intended to protect consumers, these measures can create a compliance moat, a significant barrier to entry where compliance costs become a competitive weapon [165], [284]. This dynamic favors large, incumbent firms that can afford the substantial investment in certification and legal teams, potentially stifling innovation from smaller competitors and leading to market consolidation [165], [243]. These two market failures can compound each other. A push for stringent regulation to solve the market for lemons problem could inadvertently create a formidable compliance moat, trading one market distortion for another [281], [165], [15]. An effective governance strategy must therefore include mechanisms that allow compliance obligations to scale with an organization's resources and maturity, preventing safety regulations from becoming an insurmountable barrier to new market entrants.

6.3 Geopolitical Risks Exacerbated by the Verification Gap

The verification gap exacerbates a critical geopolitical and economic vulnerability, the

concentration of the AI supply chain [253]. The dependence of the global economy on a small number of frontier foundation models, developed primarily by a few corporations in a single country, creates a systemic risk [1], [29], [253]. A single, widespread flaw in a dominant model or a malicious function inserted through a supply chain compromise could trigger cascading failures across the economies of allied nations [253].

This concentration also raises issues of data provenance and sovereignty [285]. Since these models are trained on vast datasets that may reflect the cultural biases of their origin, their global deployment can perpetuate those biases and create a dependency on a foreign data infrastructure [285], [286], [287]. This dynamic, sometimes described as a form of data colonialism, is made more acute by the verification gap [285], [286], [287]. The technical inability of a downstream user to audit a third-party model for these embedded data biases means they must either trust the vendor's claims or accept the risk of deploying a system that contains unexamined and potentially harmful assumptions [53].

This risk extends beyond the imposition of cultural values to the normalization of factual inaccuracies, connecting the technical verification gap to the broader societal crisis of verifiability [11], [53], [288]. A model trained on data reflecting a widespread public misconception will reproduce that error with statistical confidence, a phenomenon that can be described as an AI Mandela Effect in which the model delivers an answer statistically synthesized from public perception [289]. This makes it more difficult for users to question or verify certain types of information, creating a technical mechanism that can launder and legitimize misinformation at scale [290], [291]. While interventions such as heuristics, behavioral analysis and high quality training data can reduce this problem, the verification gap means that we can never be certain that this problem as been completely eliminated [141], [145], [53].

6.4 Comparative Analysis of Global Governance Approaches

The economic challenges of uncertainty and market failure create a common set of problems for global policymakers. The solutions they pursue, however, are not shaped by economic logic alone. A country's response is also deeply influenced by its unique national culture, established legal traditions, and broader strategic priorities [252]. Factors such as a nation's geopolitical standing and its ambition to achieve technical or business leadership in AI inevitably guide the governance frameworks that emerge [252]. This results in a fragmented global landscape where different jurisdictions approach the same set of problems with their own distinct assumptions and goals [252].

This has created a global tension between nations prioritizing rapid innovation, sometimes seen

as a race to the bottom on safety standards, and those aiming to establish stringent rules to gain a competitive advantage in safe and trustworthy AI, often termed "regulatory leadership" [292], [293]. The European Union often frames uncertainty through a risk-averse approach that tempers technology deployment to ensure citizen protection [294], [295], [296]. The United States exhibits a pattern of reactive regulation that often prioritizes innovation and addresses harms through enforcement actions and litigation [122], [176], [297]. China employs a state-driven model that integrates verification into command and control structures, using mandated reporting and state-supervised audits to ensure that AI development serves its strategic national priorities [298], [299], [300]. India generally emphasizes government-built digital platforms and a pilot first, iterative approach to deploying and governing AI, coordinating closely with private AI providers to scale capability while managing risk [301], [302], [303], [304]. Singapore often provides voluntary process attestation tools for key sectors, reflecting its role as a global financial hub [305], [306]. These national tendencies help explain why each model responds differently to the persistent uncertainty of AI behavior.

6.4.1 The European Union: A Rights-Based Regime

The European Union has enacted a rights based regime in the AI Act that classifies systems by risk and requires ex ante conformity assessments for systems designated as high risk before they enter the market [38]. The Act requires providers to put in place development governance measures, to produce detailed technical documentation, and to maintain continuous post market monitoring for systems designated as high risk [38]. It establishes the European Artificial Intelligence Board to coordinate implementation and harmonize enforcement guidance across member states, and it requires each member state to designate national competent authorities for market surveillance and for protection of fundamental rights [38], [307]. Firms outside the EU have signaled changes in design and reporting practices to retain access to the EU market, illustrating that EU regulation can influence global product design and reporting practices [308], [309]. By embedding conformity requirements and testing obligations at market entry, the European model makes technical assurance and documented verification prerequisites for many classes of commercial deployment [38].

6.4.2 The United States: A Sector-Specific Framework

In the United States, AI oversight has evolved through a mix of voluntary federal guidance, agency directives, and a growing patchwork of state laws, producing a decentralized regulatory landscape that emphasizes sectoral enforcement and voluntary risk management frameworks

[122]. Federal technical guidance, notably NIST's AI Risk Management Framework AI RMF 1.0 and companion materials, promotes consistent risk governance practices while remaining explicitly voluntary [56]. The White House's current executive actions focus on removing regulatory barriers to accelerate AI development and agency adoption, rather than coordinating use and procurement standards across the government [310]. Several states have enacted or proposed laws requiring transparency, reporting, or impact assessments for certain automated decision systems, producing varied state level obligations for deployers [311], [312], [313], [314]. Firms and stakeholders report that navigating this multi jurisdictional landscape creates material compliance burdens and uncertainty for smaller developers and startups [122], [315], [316]. In the absence of a single, uniform federal ex ante audit mandate, verification and accountability for AI systems in the United States typically proceed through agency enforcement, procurement conditions, sectoral rules, and litigation, which raises evidentiary and doctrinal challenges for courts and regulators when dealing with opaque systems [122], [206], [312], [315].

6.4.3 China: A State-Driven, Strategic Approach

China has taken a state driven approach, as reflected in the State Council's August 2025 Opinions on deep implementation of the AI Plus initiative [298]. The Ministry of Foreign Affairs published an AI capacity building action plan in September 2024 oriented to international capacity building and cooperation [299]. In 2023 authorities published interim measures and draft rules on generative AI services that include requirements for labeling, transparency reporting, content governance, and oversight expectations for designated services and pilot sectors [300]. Selected national champion firms receive state support for sovereign compute initiatives, model evaluation facilities, and government-linked integrated testing environments [298], [317]. Mandated reporting channels require companies to submit safety and risk information to designated authorities, and licensed audit or testing entities operate under government supervision in specified programs and pilot environments [300], [318]. Verification in China is integrated into command and control structures that serve strategic planning priorities, rather than functioning as an independent guarantee of safety [298], [319].

6.4.4 India: A Public-Private Ecosystem Approach

India combines a large engineering talent base and a growing technology market with a strategy that prioritizes government led digital public platforms and mission style programs to accelerate AI adoption while managing risks [302]. The IndiaAI Mission is building a national ecosystem

focused on compute capacity, data platforms, application development, skills, and a Safe and Trusted AI pillar through public private partnerships and dedicated mission pillars [301], [320]. The mission explicitly identifies compute and data platforms and has supported pilot projects, toolkits, and expressions of interest aimed at operationalizing bias mitigation, explainability, auditing, and other trust practices [301], [302]. Government and industry materials emphasize open standards, public digital building blocks, capacity building, and voluntary industry risk frameworks as complementary instruments to statutory controls such as the Digital Personal Data Protection Act [302], [320]. Government reports and program evaluations note continuing challenges in data standardization, workforce readiness, and operationalizing model level accountability, indicating the need for clearer processes for verification, fault attribution, and liability as India scales AI across public and private sectors [301], [302].

6.4.5 Singapore: A Market-Driven, Adopter-Focused Model

Singapore's policy emphasis is on secure, efficient adoption and governance for sectoral uses rather than on an explicit national program to build foundational models [306], [321]. Its role as a global financial center has driven rapid adoption of AI in banking and insurance and growing use in logistics [321], [306]. Singapore's approach builds on a layered governance model that integrates technical validation and risk controls at each stage of the AI lifecycle to tackle bias, adversarial manipulation, unauthorized data use, and accountability concerns [306], [322]. This approach adapts banks' existing validation frameworks and recommends bias detection tests, adversarial robustness assessments, data lineage documentation, and continuous performance monitoring to address verification gaps in opaque systems [306]. It also calls for audit trails and governance evidence to support accountability and supervisory review under current legal regimes [306]. The Model AI Governance Framework and the Generative AI Evaluation Sandbox apply those principles through sectoral pilots and evaluation exercises designed to surface harms and refine metrics before wider deployment [323]. The AI Verify toolkit provides standardized tests and machine-readable reports that capture fairness, explainability, and robustness metrics to support comparability, third party review, and contractual or regulatory checks [324], [325]. High public trust in institutions and a strong risk management culture in finance make process based assurances and iterative attestations widely adopted instruments for many firms in Singapore's financial sector [326], [327]. By embedding a cycle of continuous certification and monitoring into operations, Singapore reduces uncertainty about AI behavior in a way that aligns with its focus on practical, user oriented adoption [306], [323]. This market driven model serves as a pragmatic template that other adopter focused jurisdictions can adapt to manage AI risk while preserving innovation [328], [321].

Table 4: A Comparative Analysis of Global AI Governance Approaches

Core Philosophy

- **United States:** Fragmented federalism with sectoral regulation and uneven political momentum. Tends to prioritize innovation and market leadership rather than a unified national strategy.
- **European Union:** Human centric and rights based. Proactively legislates risk based controls and seeks to shape international norms through regulatory leadership (the "Brussels Effect").
- **China:** State driven and stability focused. Prioritizes digital sovereignty and state control over the AI supply chain.
- **India:** Pragmatic and development focused. Prioritizes digital sovereignty and capacity building through programs like IndiaAI and emphasizes open source and home-grown solutions as strategic assets.
- **United Kingdom:** Pro innovation and sector specific. Balances attracting investment with governance through existing regulators and positions itself as an attractive jurisdiction for safety related innovation.
- **Singapore:** Pragmatic and market enabling. Provides voluntary, government backed toolkits and sandboxes to help industry manage uncertainty and demonstrate trustworthiness.

Regulatory Model

- **United States:** Sector specific and largely ex post enforcement. A patchwork of state laws and sectoral federal rules produces uneven coverage in the absence of a comprehensive national statute.
- **European Union:** Comprehensive, horizontal and risk based under the AI Act. Uses ex ante conformity assessments for systems classified as high risk.
- **China:** Top down and state directed. A rapidly evolving mix of statutory, regulatory, and administrative measures aligned with state strategic priorities.
- **India:** Adaptive and consultative. Combines multi stakeholder drafts, legal gap analysis, and process based compliance with active debate over AI specific statutes versus amendments to existing laws.
- **United Kingdom:** Sector specific and principles based. Relies on existing regulators supported by central coordination, and proposals for a statutory AI Authority remain under discussion.
- **Singapore:** Co-regulatory and largely voluntary. Relies on soft law and guidance, and on government provided toolkits and sandboxes to enable safe adoption.

Liability Approach

- **United States:** Relies primarily on traditional tort law and product liability doctrines, which face evidentiary challenges created by model opacity and verification gaps.
- **European Union:** Moves toward expanded producer liability. Revisions to the Product Liability Directive increase strict liability exposure for certain AI related harms.
- **China:** Liability framework is developing. Enforcement and compliance are embedded within state supervisory and administrative processes that prioritize state objectives.
- **India:** Exploring doctrinal reform to address bias, attribution and autonomous system liability. Relies on common law and IT Act provisions for now, while experts call for clearer civil liability standards to address verification issues.
- **United Kingdom:** Relies on existing common law principles and sectoral regulatory tools, with ongoing discussion about whether targeted statutory reforms are needed.
- **Singapore:** Relies on existing common law principles. Regulatory guidance and sector specific expectations from MAS complement common law remedies rather than replacing them.

7. Conclusion: A Framework for Governing Under Uncertainty

Sections 2 and 3 show that black box behavioral tests and mechanistic interpretability are insufficient on their own to close the verification gap [8], [16]. Section 4 reframes regulation around process governance as a tractable basis for accountability when direct model audits are insufficient [56], [57]. Section 5 documents how the inability to verify model internals erects an evidentiary barrier that implicates negligence and product liability doctrines, and complicates fault attribution when internal process failures cause external harm [41], [42]. Section 6 compares how major developer and adopter economies confront persistent uncertainty under opaque AI systems [252]. Throughout this analysis, the paper has shown how this technical gap creates not only legal and economic uncertainty but also novel organizational challenges to human oversight and deepens systemic risks within the AI supply chain [2], [29], [33]. Together, these findings indicate that policymakers should plan for persistent verification limits implied by current theoretical and empirical work, and prioritize governance approaches that do not depend on perfect verification [8], [16]. This analysis has argued that a shift to process-based governance is not simply a preference, but a necessary consequence of the formal limits of technical verification.

The verification gap generates a clear market paradox, but its effects are broader than they may first appear [1]. The barrier to adoption is not limited to risk-averse enterprises in heavily

regulated sectors [2]. Even companies deploying AI for seemingly low-stakes internal tasks, such as software development or data analysis, inherit a new and poorly understood category of operational and cybersecurity risk [31]. As this paper has shown, these internal process risks can become the source of future external liabilities [42]. A subtle, AI-generated flaw in a product's code can evade standard quality assurance and lead to a catastrophic failure, creating an evidentiary challenge for both the company and any injured parties [31], [41]. The central barrier to safe and scalable AI integration is therefore not just legal and regulatory uncertainty, but a more fundamental business uncertainty about the reliability and security of an increasingly critical production tool [2].

This paper proposes a three-pillar strategy for a realistic and pragmatic roadmap, with a common focus on enabling substantive human oversight, which requires processes designed to counter automation bias and ensure that human judgment serves as a genuine check on AI systems [32], [33]. These pillars are not independent solutions, but a self-reinforcing system designed to address the distinct layers of the verification gap. A tiered liability framework and process-based governance are designed to directly close the asymmetric gap. They create powerful legal and economic incentives for developers to share information and demonstrate diligence, making the internal process an auditable basis for accountability. This is essential because technical solutions alone cannot succeed. Sustained foundational research, the third pillar, is aimed at shrinking the practical gap by developing more effective and scalable auditing tools. Both of these efforts are made necessary by the existence of the foundational gap and the observational gap. The permanence of computational intractability and the inherent unpredictability of emergent model behavior suggest that technical solutions alone may not fully resolve this problem [16], [19]. This highlights the need to build robust governance frameworks to manage the risks that are likely to remain.

7.1 A Roadmap for Governing Under Uncertainty

The core of the information asymmetry lies in the developer's exclusive control over the initial design process [16]. The following three pillars are therefore aimed directly at creating incentives and accountability for those crucial, upstream choices. These pillars are designed as a self-reinforcing system precisely to counter the risk of compliance theater. First, the analysis suggests that policymakers could prioritize the bargained-for tiered liability framework detailed in Section 5. This framework acts as the engine of the entire system. It creates powerful economic and legal incentives that make superficial compliance a high-risk business strategy by establishing distinct, risk-appropriate liability defaults and safe harbors for non-commercial, general commercial, and high-risk applications. The threat of defaulting to a stricter liability standard or losing a statutory safe harbor provides a compelling reason for developers to engage

with process governance genuinely. While other market and regulatory incentives can encourage the adoption of process standards, this liability framework provides the critical, non-negotiable consequences necessary to ensure that such governance is consistently substantive. Its central feature is an administrative pathway that allows developers to earn a more predictable and less demanding legal standard in exchange for deep transparency with an expert regulator, providing legal certainty for industry and promoting the adoption of auditable safety processes.

Second, the findings support establishing process-based governance as a foundational pillar of oversight. This approach complements direct model auditing by providing the necessary context to make those audits meaningful and by creating a separate, more tractable basis for accountability. Since technical and behavioral audits alone are insufficient, a verifiable process provides a separate, more tractable basis for accountability for external regulators, internal risk managers, and enterprise customers seeking to verify the integrity of their AI supply chain [8], [16], [57]. Such processes are also essential for structuring the human-AI interaction in a way that promotes effective oversight rather than allowing it to become a procedural formality [33], [329]. This can be achieved through a dual-track implementation path. The public sector can lead by formally adopting the NIST AI RMF for all federal agencies and contractors, supplemented with mandatory implementation standards [56]. Concurrently, private sector adoption of ISO/IEC 42001 certification can be encouraged by linking it to legal benefits, such as its use in demonstrating a reasonable standard of care [147], [197, Sec. 13. Custom], [160, Sec. 908(2)].

Third, the findings support a sustained commitment to foundational research, framed not as a search for a technical silver bullet that would replace process governance, but as a means of supporting it. The most pressing research agenda is not to solve intractability, but to make the human-led process itself more reliable and auditable [16]. Funding should therefore prioritize two main areas. The first is the development of tools and standards for creating more reliable and less burdensome evidentiary trails for AI development. The second is research into methods and interfaces that enhance the efficacy of human oversight by countering the effects of automation bias and improving the cognitive performance of human reviewers.

7.2 Deepening the Policy Toolkit: The AI Liability Sandbox

To mitigate the economic costs of new liability regimes for startups, policymakers should explore the creation of an AI Liability Sandbox. Regulatory sandboxes provide a controlled environment for testing innovative technologies under a regulator's supervision [330]. Citing the framework proposed by The Datasphere Initiative, a concrete design would involve offering startups a time-limited safe harbor from certain liability claims [330]. Such a program would be designed with a dual purpose. It would lower the barrier to entry for new firms, while simultaneously enabling "regulatory learning" that makes the entire governance ecosystem more

efficient and evidence-based.

For participating startups, a sandbox could offer a time-limited safe harbor from certain liability claims. In exchange, these firms would engage in mandatory, confidential data sharing with regulators on safety incidents, model performance, and internal risk mitigation strategies [330]. Crucially, oversight within the sandbox would follow a principle of graduated compliance. While all participants would be held to the core safety and process principles of the governing framework, the expectations for administrative structure and documentation would be scaled to the organization's size and resources. This approach, which has precedent in other regulated sectors, would allow smaller entities to demonstrate diligence without facing the prohibitive overhead required of large, mature corporations [154].

The primary output of the sandbox would be regulatory learning. By analyzing the aggregated and anonymized incident and performance data, policymakers could develop an empirical basis for crafting more targeted regulations for the entire industry. This process would allow governance to evolve based on real-world data, moving beyond theoretical debates. The insights gained would inform the creation of streamlined guidance documents, standardized templates for process artifacts, and potentially more efficient pathways to achieving a statutory safe harbor. A sandbox thus becomes a dynamic tool for managing the inherent tension between safety and innovation, ensuring that regulation does not unduly hamper competition.

7.3 Future Research Directions: From Theory to Practice

The research agenda that emerges from this paper's analysis is pragmatic and aligned with its process-based thesis. It prioritizes the development of tools that make human-led governance more robust, reliable, and efficient. To illustrate the kinds of forward-looking concepts that fall within this agenda, one can consider several areas. For example, research into cryptographic assurance could explore methods like Zero-Knowledge Proofs to allow a developer to prove a model has certain properties without revealing proprietary information [331]. Similarly, research into confidential computing could advance the use of Trusted Execution Environments (TEEs) for runtime monitoring, creating verifiable logs of a model's behavior post-deployment [129], [130].

These specific technologies, while promising, remain speculative. Their inclusion here is not prescriptive, but illustrative of a broader principle. The most fruitful research directions are those that provide new tools for verifying process integrity and enhancing human judgment. By focusing on these goals, the research community can shrink the practical and observational gaps, reinforcing the entire governance framework. The path outlined in this paper is demanding. The implementation of a substantive audit regime and a tiered liability framework presents significant

challenges. The central argument, however, is that the verification gap reframes the problem of accountability. By demonstrating that a purely technical assurance paradigm is insufficient, this analysis argues for a complementary shift to focusing on the more tractable human-led development process. It recasts the residual challenge as a difficult but manageable forensic and evidentiary one, allowing legal and regulatory systems to engage with the problem using a more established, if still imperfect, toolkit.

Appendix: Decision-Maker's Cheat Sheet

Table 5: A Practitioner's Roadmap to AI Governance Under Uncertainty

1. Tiered Liability Framework

- **Key Challenge:** The Evidentiary Gap. Proving causation is intractable when an opaque model causes harm.
- **Core Recommendation:** Align Liability with Risk & Capacity. Implement a tiered system that creates market incentives for safety without stifling non-commercial innovation.
- **For Legal & Policy Teams:**
    - Draft legislation for a rebuttable presumption of causality.
    - Establish strict liability for designated high-risk applications.
    - Create a robust safe harbor for non-commercial research.
    - Use AI Liability Sandboxes to enable regulatory learning and build institutional capacity.
- **For VCs & Business Leaders:**
    - Price risk using new instruments like parametric insurance for AI failures, with premiums tied to process adherence.
    - In due diligence, demand evidence of auditable processes to assess the integrity of a vendor's AI supply chain.
- **For Technical & Research Teams:**
    - Develop objective, verifiable metrics of model performance that can serve as triggers for parametric insurance policies.
    - Create technical standards for AI supply chain transparency, like model bills of materials.

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2. Process-Based Governance

- **Key Challenge:** Compliance Theater. Voluntary frameworks and opaque models allow for superficial adherence to rules without substantive risk mitigation.
- **Core Recommendation:** Make Human Accountability Auditable. Shift regulatory focus from the intractable model to the tractable human-led process that created it.
- **For Legal & Policy Teams:**
    - Leverage existing authority (e.g., FTC Section 5) to define failure to follow

process standards as an "unfair practice."
    - Mandate recognized certifications (e.g., ISO 42001) as a prerequisite for liability safe harbors.
- **For VCs & Business Leaders:**
    - Adopt process standards for both compliance and operational resilience.
    - Require portfolio companies to produce auditable artifacts like Model Cards and Datasheets.
    - Design human oversight protocols that ensure substantive review and counter automation bias.
- **For Technical & Research Teams:**
    - Implement rigorous internal documentation and version control for models, data, and testing protocols.
    - Automate the generation of artifacts required for process audits.
    - Build interfaces that support meaningful human oversight.

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3. Hardening the Audit Process

- **Key Challenge:** Sophisticated Fraud. The value of certifications and safe harbors creates a powerful incentive for systemic, fraudulent compliance.
- **Core Recommendation:** Make Deception Economically Irrational. Implement an adaptive enforcement regime with severe consequences for discovered fraud.
- **For Legal & Policy Teams:**
    - Establish an adaptive re-audit system triggered by risk signals.
    - Institute severe penalties for discovered fraud, including certification revocation, criminal referrals, and punitive financial damages.
- **For VCs & Business Leaders:**
    - Contractually require audit partners to consent to potential re-audits of both internal systems and key vendors.
    - Establish and promote confidential whistleblower channels for reporting internal compliance failures.
- **For Technical & Research Teams:**
    - Develop automated tools for monitoring deployed model performance against test benchmarks to flag anomalous deviations that could trigger a re-audit.

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4. Foundational Research

- **Key Challenge:** Supporting Process Governance. Research must provide tools to make human-led processes more reliable, not chase technical "silver bullets."
- **Core Recommendation:** Invest in Tractable Oversight Tools. Acknowledge intractability and redirect research toward tools that support the process-based framework.
- **For Legal & Policy Teams:**
    - Create funding calls for research into: reliable and efficient evidentiary trails, and tools that enhance the efficacy of human oversight.
- **For VCs & Business Leaders:**
    - Partner with academic institutions on pilot projects in process instrumentation and human-computer interaction for AI oversight.
- **For Technical & Research Teams:**
    - Develop standards for auditable AI development logs and data provenance.
    - Design and test interfaces that are demonstrated to counter automation bias in human reviewers.

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5. Nuanced Technical Acknowledgment

- **Key Challenge:** The Governance Gap. Relying solely on process governance without any technical inspection of the model is insufficient, while relying solely on technical inspection is impossible.
- **Core Recommendation:** Use Heuristics as a Foundational Layer. Integrate practical, heuristic-based auditing as a necessary but insufficient component of a larger, process-based assurance framework.
- **For Legal & Policy Teams:**
    - Draft rules that treat evidence of rigorous heuristic auditing as a necessary component of demonstrating a reasonable standard of care, but not as a sufficient condition for a safe harbor in high-risk contexts.
- **For VCs & Business Leaders:**
    - Mandate heuristic auditing of third-party models as a standard part of technical due diligence to identify known vulnerabilities.
    - Understand that these audits must supplement, not replace, rigorous diligence into a vendor’s process integrity and data provenance.
- **For Technical & Research Teams:**
    - Continue to advance heuristic circuit discovery methods to improve their coverage and efficiency.

- Focus research on developing methods capable of flagging novel or anomalous mechanisms that fall outside the scope of known circuit patterns.

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